fields when they are present in the client certificate used by the Istio sidecar proxy for mTLS. Istio's own certificates use a URI SAN entry to identify clients and servers, using a standard called *SPIFFE* (Secure Production Identity Framework for Everyone), which provides a way to name services in microservices environments. Figure 11.6 shows the components of a SPIFFE identifier, which consists of a trust domain and a path. In Istio, the workload identifier consists of the Kubernetes namespace and service account. SPIFFE allows Kubernetes services to be given stable IDs that can be included in a certificate without having to publish DNS entries for each one; Istio can use its knowledge of Kubernetes metadata to ensure that the SPIFFE ID matches the service a client is connecting to.

> **DEFINITION *SPIFFE*** stands for *Secure Production Identity Framework for Everyone* and is a standard URI for identifying services and workloads running in a cluster. See https://spiffe.io for more information.

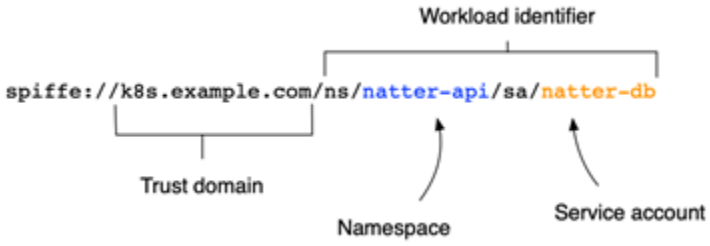


Figure 11.6 A SPIFFE identifier consists of a trust domain and a workload identifier. In Istio, the workload identifier is made up of the namespace and service account of the service.